SR UNIVERSITY

School of Computer Science and Artificial Intelligence

Lab Assignment # 14

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Batch No. : 22

Task Description 1 – (Landing Page Design)
• Create a simple landing page for a product or service. Include a header, hero section, and footer.
Requirements:
• Use HTML and CSS to structure the page.
• Use AI to suggest layout alignment and typography.
• Ensure the page is responsive across devices.

Prompt:-
Create a simple landing page for a product or service using **HTML and CSS**. The page should include a **header with navigation, a hero section with title and description, and a footer**. Make the layout clean and ensure the page is **responsive on mobile, tablet, and desktop devices**.
Code:-

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```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,initial-scale=1">
    <title>CloudSync</title>
    <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.5.0/all.min.css">
    <style>
        *{margin:0;padding:0;box-sizing:border-box}
        :root{--p:#00c;--s:#ccc;--d:#111;--l:#fefefe;--g:#666;--w:#fff}
        html{scroll-behavior:smooth}
        body{font-family:Segoe UI;line-height:1.5;color:var(--g);background:var(--l)}
        section{padding:80px 10px}
        .max{max-width:1200px;margin:0 auto}
        header{background:var(--w);box-shadow:0 1px 10px rgba(0,0,0,.1);position:sticky;top:0;z-index:100}
        header>div{display:flex;justify-content:space-between;align-items:center;padding:15px}
        .logo{font-size:24px;font-weight:700;color:var(--p);display:flex;gap:8px}
        .nav{display:flex;gap:30px;list-style:none}
        a{text-decoration:none;color:var(--g)}.nav a{padding:8px 0;border-bottom:2px solid transparent}
        .nav a:hover{color:var(--p);border-bottom-color:var(--p)}
        .hero{background:linear-gradient(135deg,var(--p),var(--s));color:var(--w);padding:100px 30px;text-align:center;position:relative;overflow:hidden}
        h1{font-size:48px;margin:0 0 20px;position:relative;z-index:1}
        .hero p{font-size:18px;opacity:.95;margin-bottom:30px;position:relative;z-index:1}
        .hero i{font-size:500px;opacity:.1;position:absolute;right:-100px;top:50%;transform:translateY(-50%)}
        .btn{background:var(--w);color:var(--p);padding:12px 35px;border:none;border-radius:5px;font-weight:600;cursor:pointer;transition:.3s}
        .btn:hover{transform:translateY(-2px);box-shadow:0 5px 20px rgba(0,0,0,.2)}
        .hero .btn{box-shadow:0 2px 10px rgba(0,0,0,.1)}
        h2{font-size:42px;color:var(--d);text-align:center;margin-bottom:50px}
        .grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(260px,1fr));gap:40px;margin:0px 0}
        .card{background:var(--l);padding:30px;border-radius:8px;text-align:center;transition:.3s;box-shadow:0 2px 5px rgba(0,0,0,.05)}
        .card:hover{transform:translateY(-5px);box-shadow:0 10px 30px rgba(0,0,0,.1)}
        .icon{font-size:48px;color:var(--p);margin-bottom:20px}
        h3{font-size:21px;color:var(--d);margin-bottom:10px}
        .features{background:var(--w)}
        .pricing{background:var(--l)}
        .card.pro{border-top-color:var(--p);transform:scale(1.05)}
        .price{font-size:42px;color:var(--p);margin:20px 0}
        .list{list-style:none;text-align:left;margin:0 0}
        .list li{padding:10px 0;border-bottom:1px solid #eee}
        .pro{width:100%;padding:12px;border:2px solid var(--p);background:var(--w);color:var(--p);font-weight:600;border-radius:5px;cursor:pointer;transition:.3s;margin-top:20px}
        .pro:hover{background:var(--p);color:var(--w)}
        .contact{background:var(--w)}
        .form{max-width:600px;margin:0 auto}
        input,textarea{width:100%;padding:15px;border:1px solid #ddd;border-radius:5px;margin:10px 0;font-family:inherit;font-size:16px}
        input:focus,textarea:focus{outline:0;border-color:var(--p);box-shadow:0 0 5px rgba(0,102,204,.1)}
        .footer{background:var(--d);color:var(--w);padding:30px 20px;text-align:center;font-size:16px}
        .social{display:flex;gap:15px;justify-content:center;margin-top:15px}
        .social a{display:flex;align-items:center;justify-content:center;width:40px;height:40px;background:rgba(255,255,255,.1);color:var(--w);border-radius:50%;transition:.3s}
        .social a:hover{background:var(--p)}
        @media(max-width:768px){.nav{gap:15px;font-size:13px}.hero{padding:60px 20px}h1{font-size:36px}h2{font-size:32px}.grid{grid-template-columns:1fr}.card.pro{transform:scale(1)}}
        @media(max-width:480px){.logo{font-size:20px}.nav{gap:10px}.hero{padding:40px 15px}h1{font-size:28px}h2{font-size:24px}h3{font-size:18px}.icon{font-size:40px}.price{font-size:32px}.btn{padding:10px 20px}}
    </style>
</head>
<body>
    <header>
        <div class="nav">
            <div class="logo">🌐 CloudSync</div>
            <nav class="nav">
                <a href="#">Home</a>
                <a href="#">Features</a>
                <a href="#">Contact</a>
            </nav>
        </div>
    </header>
```

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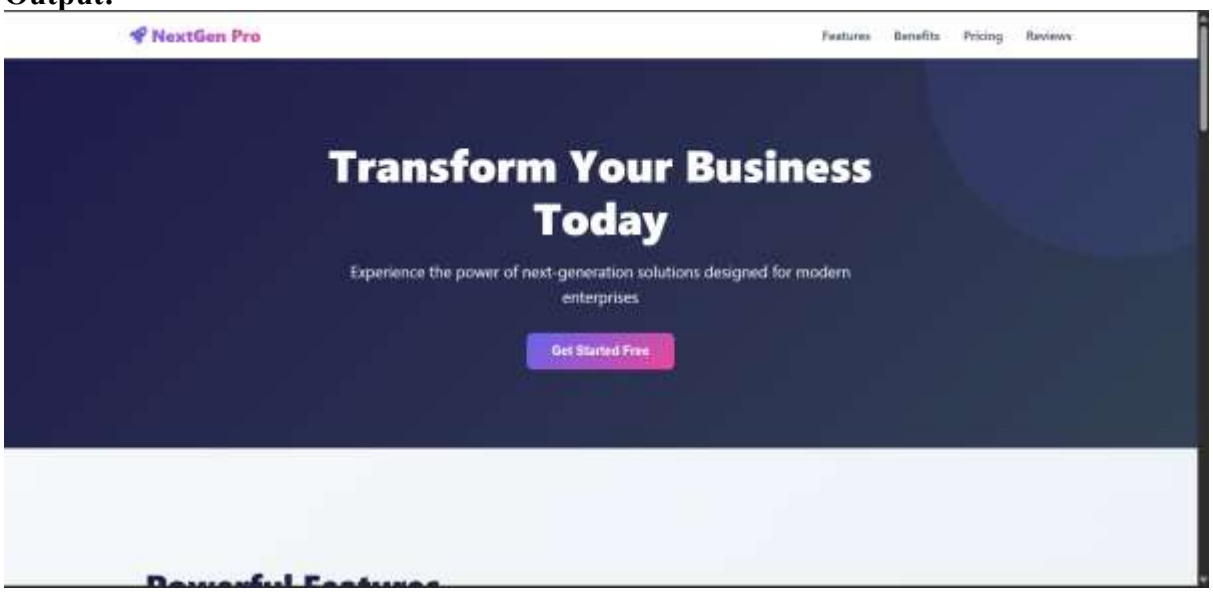
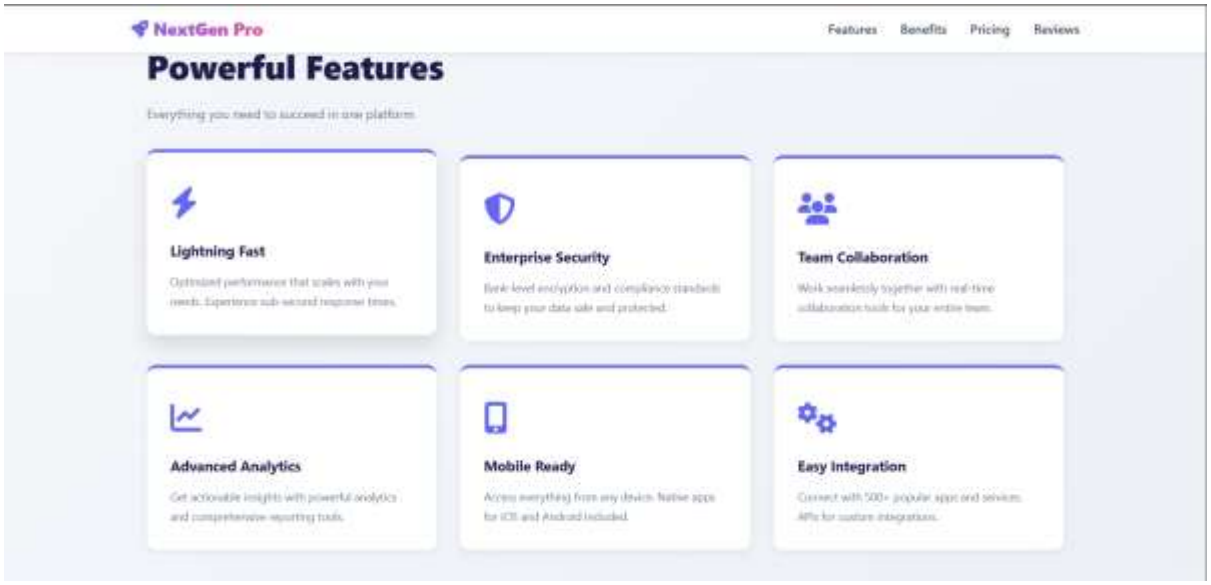
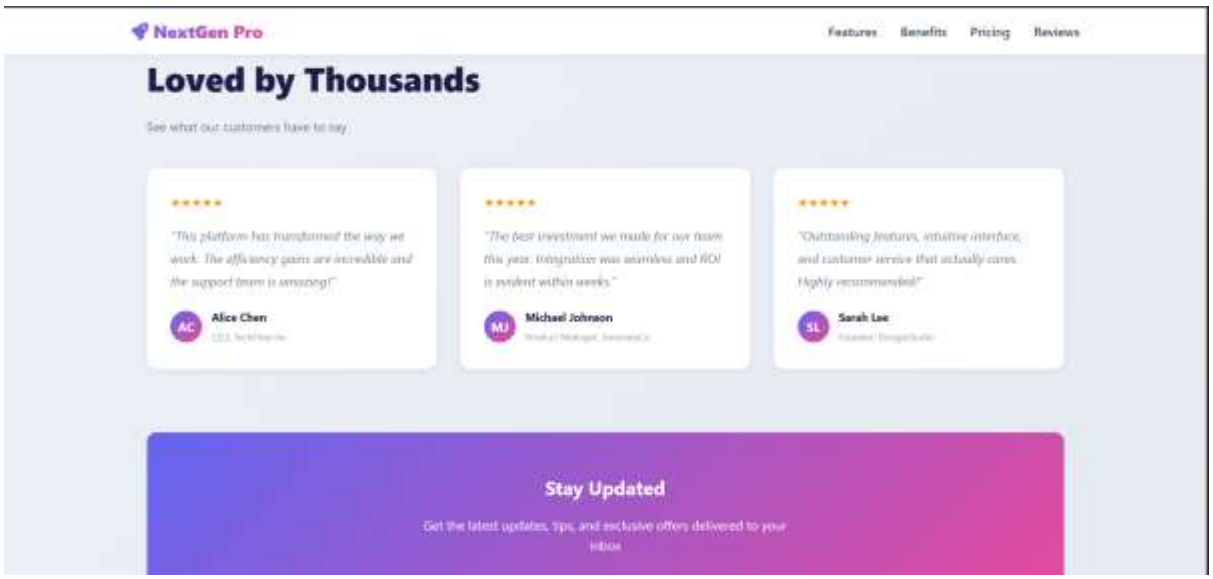
```html
<body>
    <header>
    </header>
    <section class="hero" id="h">
        <div class="max">
            <h1>Your Files, Anywhere</h1>
            <p>Secure cloud storage for all your devices</p>
            <button class="btn">Get Started</button>
        </div>
        <i class="fas fa-cloud-upload-alt"></i>
    </section>
    <section class="features" id="f">
        <div class="max">
            <h2>Why CloudSync?</h2>
            <div class="grid">
                <div class="card"><div class="icon"><i class="fas fa-lock"></i></div><h3>Secure</h3></div>
                <div class="card"><div class="icon"><i class="fas fa-bolt"></i></div><h3>Fast</h3></div>
                <div class="card"><div class="icon"><i class="fas fa-share"></i></div><h3>Share</h3></div>
                <div class="card"><div class="icon"><i class="fas fa-mobile"></i></div><h3>Mobile</h3></div>
                <div class="card"><div class="icon"><i class="fas fa-sync"></i></div><h3>Sync</h3></div>
                <div class="card"><div class="icon"><i class="fas fa-headset"></i></div><h3>Support</h3></div>
            </div>
        </div>
    </section>
    <section class="pricing">
        <div class="max">
            <h2>Pricing</h2>
            <div class="grid">
                <div class="pcard">
                    <h3>Basic</h3><p class="price">$0/mo</p>
                    <ul class="list">
                        <li>✓ 5 GB Storage</li>
                        <li>✓ Basic Support</li>
                        <li>✓ 1 Device</li>
                    </ul>
                    <button class="pbtn">Free</button>
                </div>
                <div class="pcard pro">
                    <h3>Pro</h3><p class="price">$9.99/mo</p>
                    <ul class="list">
                        <li>✓ 1 TB Storage</li>
                        <li>✓ Priority Support</li>
                        <li>✓ Unlimited</li>
                    </ul>
                    <button class="pbtn" style="background:var(--p);color:var(--w)">Choose</button>
                </div>
                <div class="pcard">
                    <h3>Business</h3><p class="price">$19.99/mo</p>
                    <ul class="list">
                        <li>✓ Unlimited Storage</li>
                        <li>✓ 24/7 Support</li>
                        <li>✓ Advanced Features</li>
                    </ul>
                    <button class="pbtn">Choose</button>
```

```html
                    <li>✓ Unlimited Storage</li>
                    <li>✓ 24/7 Support</li>
                    <li>✓ Advanced Features</li>
                </ul>
                <button class="pbtn">Choose</button>
            </div>
        </div>
    </div>
</section>
<section class="contact" id="c">
    <div class="cform">
        <h2>Get In Touch</h2>
        <form>
            <input type="text" placeholder="Name" required>
            <input type="email" placeholder="Email" required>
            <textarea placeholder="Message" rows="5" required></textarea>
            <button type="submit" class="btn">Send</button>
        </form>
    </div>
</section>
<footer class="footer">
    <div class="max">
        <p>&copy; 2024 CloudSync. All rights reserved.</p>
        <div class="social">
            <a href="#" title="Facebook"><i class="fab fa-facebook"></i></a>
            <a href="#" title="Twitter"><i class="fab fa-twitter"></i></a>
            <a href="#" title="LinkedIn"><i class="fab fa-linkedin"></i></a>
            <a href="#" title="Instagram"><i class="fab fa-instagram"></i></a>
        </div>
    </div>
</footer>
</body>
```

Output:-


NextGen Pro
Features
Benefits
Pricing
Reviews
Transform Your Business Today
Experience the power of next-generation solutions designed for modern enterprises
Get Started Free


NextGen Pro
Features
Benefits
Pricing
Reviews
Powerful Features
Lightning Fast
Enterprise Security
Team Collaboration
Advanced Analytics
Mobile Ready
Easy Integration


NextGen Pro
Features
Benefits
Pricing
Reviews
Loved by Thousands
Alice Chen
Michael Johnson
Sarah Lee
Stay Updated

Justification:-
This prompt helps generate a basic landing page with a clear structure. It focuses on proper layout, simple typography, and responsiveness so the page looks good on different screen sizes.

Task Description 2 – (Theme Toggle Feature)
• Provide a button to switch between light and dark themes..
Requirements:
• Use JavaScript to toggle CSS classes.
• Use AI assistance to manage color contrast and accessibility.
• Ensure theme preference applies instantly on click.

Prompt:-
Create a **light and dark theme toggle button** using **HTML, CSS, and JavaScript**. When the user clicks the button, the website theme should change between light and dark mode by toggling CSS classes.

Code:-

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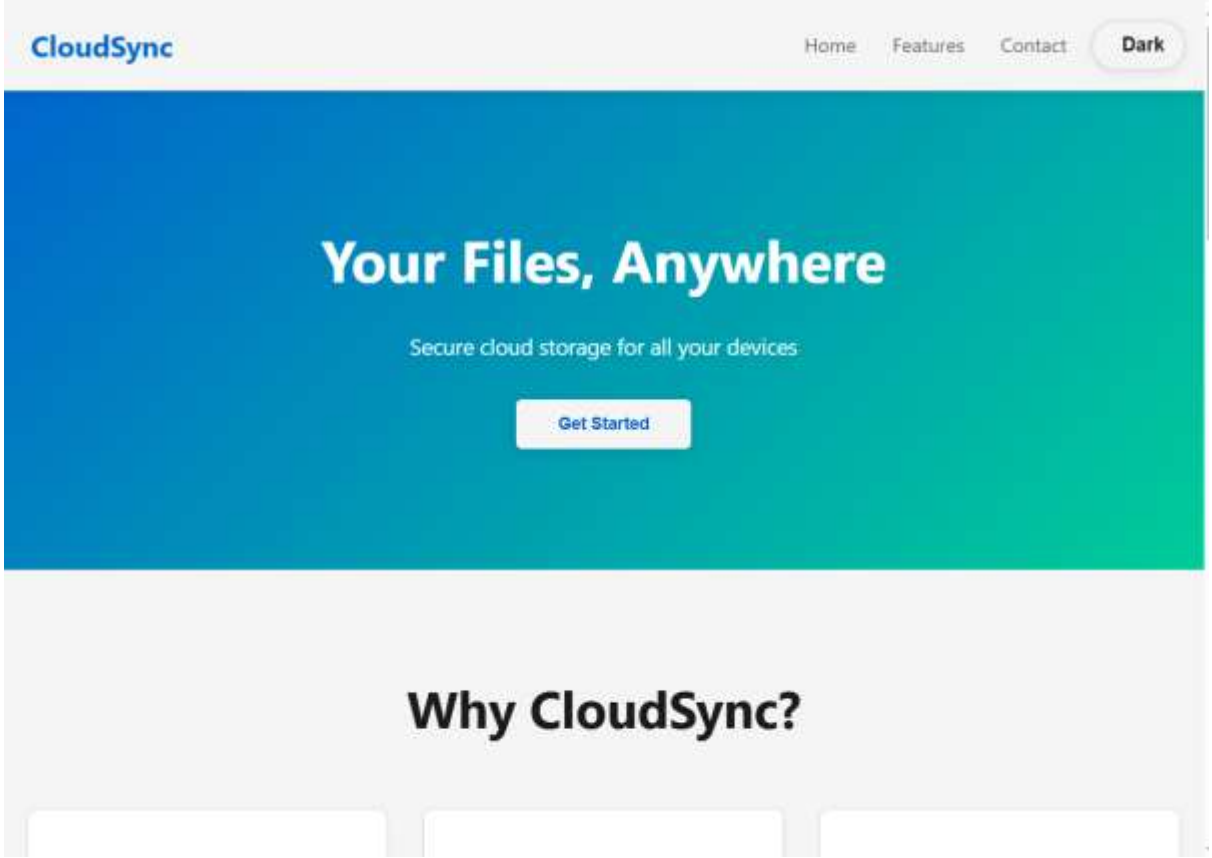
```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width,initial-scale=1">
    <title>CloudSync</title>
    <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6/css/all.min.css">
    <style>
        *{margin:0;padding:0;box-sizing:border-box}
        :root{--lb:#fff;--ls:#f5f5f5;--lt:#1a1a1a;--lm:#666;--la:#06c;--la2:#0c9;--db:#1a1a1a;--ds:#2d2d2d;--dt:#fff;--dm:
        body{background:var(--lb);color:var(--lt);font-family:Segoe UI,system-ui;line-height:1.6;transition:all 0.3s}
        body.dark-theme{background:var(--db);color:var(--dt)}
        section{padding:80px 20px}.max{max-width:1200px;margin:0 auto}header{background:var(--ls);box-shadow:0 2px 10px rgba(0,0,0,.08);posit
        body.dark-theme header{background:var(--ds);border-color:#404040}
        header>div{display:flex;justify-content:space-between;align-items:center;padding:15px}
        .logo{font-size:24px;font-weight:700;color:var(--la);display:flex;gap:8px}
        body.dark-theme .logo{color:var(--da)}
        .header-right{display:flex;align-items:center;gap:20px}
        .nav{display:flex;gap:30px;list-style:none}
        a{text-decoration:none;color:var(--lm);transition:.3s;padding:8px 0;border-bottom:2px solid transparent}
        body.dark-theme a{color:var(--dm)}
        .nav a:hover{color:var(--la);border-bottom-color:var(--la)}
        body.dark-theme .nav a:hover{color:var(--da);border-bottom-color:var(--da)}
        .theme-toggle{display:flex;align-items:center;gap:8px;padding:10px 16px;border:2px solid #e0e0e0;border-radius:25px;background:var(--
        body.dark-theme .theme-toggle{background:var(--ds);color:var(--dt);border-color:#404040}
        .theme-toggle:hover{transform:translateY(-2px);box-shadow:0 4px 16px rgba(0,102,204,.25);border-color:var(--la)}
        body.dark-theme .theme-toggle:hover{border-color:var(--da);box-shadow:0 4px 16px rgba(51,153,255,.25)}
        .hero{background:linear-gradient(135deg,var(--la),var(--la2));color:#fff;padding:100px 20px;text-align:center;position:relative;overf
        body.dark-theme .hero{background:linear-gradient(135deg,var(--da),var(--da2))}
        h1{font-size:48px;margin:0 0 20px;z-index:1;position:relative}
        .hero p{font-size:18px;opacity:.95;margin-bottom:30px;z-index:1;position:relative}
        .hero i{font-size:120px;opacity:.2;position:absolute;right:30px;top:50%;transform:translateY(-50%)}
        .btn{background:var(--ls);color:var(--la);padding:12px 35px;border:none;border-radius:5px;font-weight:600;cursor:pointer;transition:.3s
        body.dark-theme .btn{background:var(--ds);color:var(--da)}
        .btn:hover{transform:translateY(-2px);box-shadow:0 5px 20px rgba(0,102,204,.2)}
        h2{font-size:42px;color:var(--lt);text-align:center;margin-bottom:50px}
        body.dark-theme h2{color:var(--dt)}
        .features{background:var(--ls)}body.dark-theme .features{background:var(--ds)}
        .grid{display:grid;grid-template-columns:repeat(auto-fit,minmax(280px,1fr));gap:30px;margin:30px 0}
        .card{background:var(--lb);padding:30px;border-radius:8px;text-align:center;transition:.3s;box-shadow:0 2px 5px rgba(0,0,0,.05);borde
        body.dark-theme .card{background:var(--ds);box-shadow:0 2px 5px rgba(0,0,0,.3);border-color:#404040;color:var(--dt)}
        .card:hover{transform:translateY(-5px);box-shadow:0 8px 15px rgba(0,102,204,.15)}
        .icon{font-size:48px;color:var(--la);margin-bottom:20px}
        body.dark-theme .icon{color:var(--da)}
        h3{font-size:22px;color:var(--lt);margin-bottom:10px}
        body.dark-theme h3{color:var(--dt)}
        .pcard{background:var(--ls);padding:40px;border-radius:8px;border-top:4px solid #e0e0e0;text-align:center;box-shadow:0 2px 10px rgb
        body.dark-theme .pcard{background:var(--ds);border-color:#404040;box-shadow:0 2px 10px rgba(0,0,0,.3)}
        .pcard.pro{border-top-color:var(--la);transform:scale(1.05)}
        body.dark-theme .pcard.pro{border-top-color:var(--da)}
        .price{font-size:42px;color:var(--la);margin:20px 0}
        body.dark-theme .price{color:var(--da)}
        .list{list-style:none;margin:30px 0}
        .list li{padding:10px 0;border-bottom:1px solid #e0e0e0;color:var(--lm)}
```

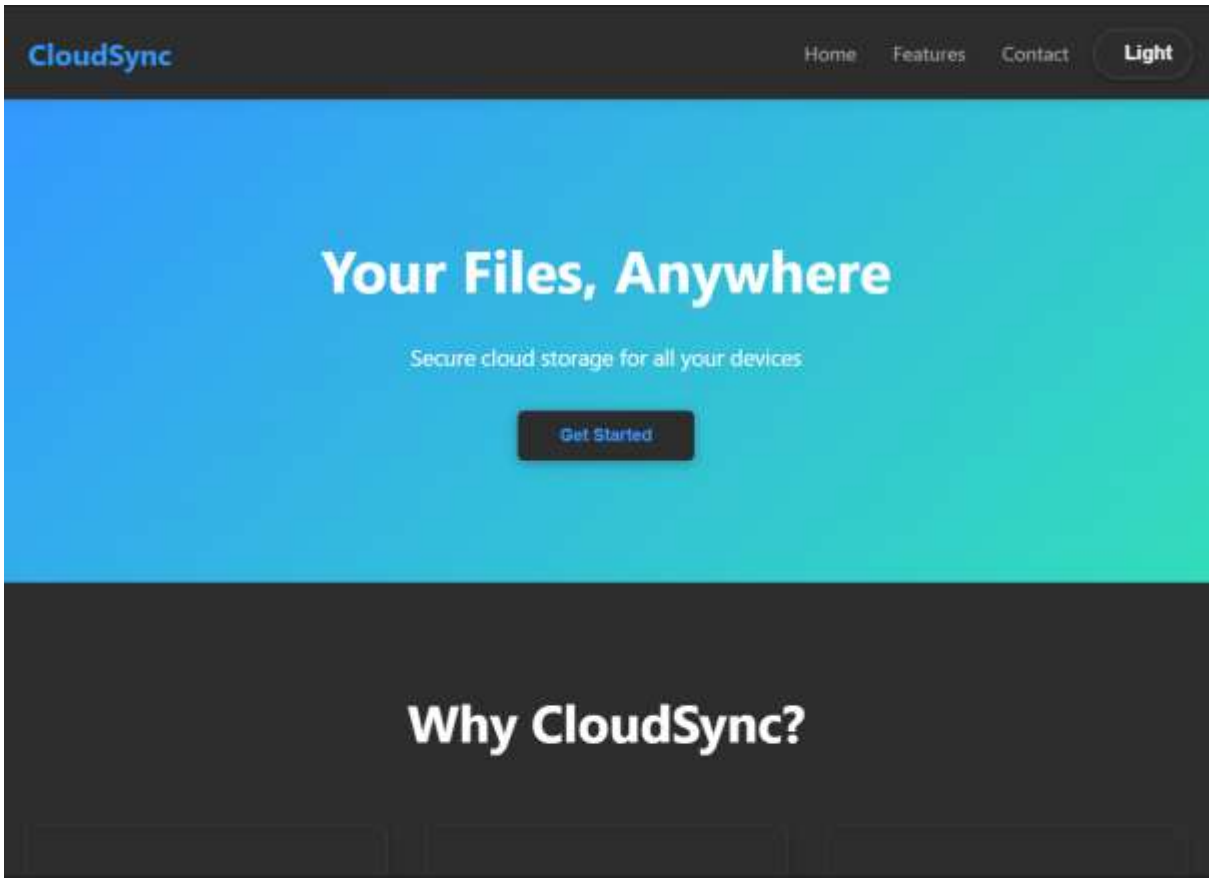
```
    .pcard.pro{border-top-color:var(--la);transform:scale(1.05)}
    body.dark-theme .pcard.pro{border-top-color:var(--da)}
    .price{font-size:42px;color:var(--la);margin:20px 0}
    body.dark-theme .price{color:var(--da)}
    .list{list-style:none;margin:30px 0}
    .list li{padding:10px 0;border-bottom:1px solid #e0e0e0;color:var(--lm)}
    body.dark-theme .list li{border-color:#404040;color:var(--dm)}
    .pbtn{width:100%;padding:12px;border:2px solid var(--la);background:transparent;color:var(--la);font-weight:600;border-radi
    body.dark-theme .pbtn{border-color:var(--da);color:var(--da)}
    .pbtn:hover{background:var(--la);color:#fff}
    body.dark-theme .pbtn:hover{background:var(--da)}
    .contact{background:var(--ls)}body.dark-theme .contact{background:var(--ds)}
    .cform{max-width:600px;margin:0 auto}
    input,textarea{width:100%;padding:12px;border:1px solid #e0e0e0;border-radius:5px;margin:10px 0;font-family:inherit;backg
    body.dark-theme input,body.dark-theme textarea{background:var(--ds);color:var(--dt);border-color:#404040}
    input:focus,textarea:focus{outline:0;border-color:var(--la);box-shadow:0 0 5px rgba(0,102,204,.2)}
    body.dark-theme input:focus,body.dark-theme textarea:focus{border-color:var(--da);box-shadow:0 0 5px rgba(51,153,255,.3)}
    footer{background:var(--lt);color:var(--ls);padding:30px 20px;text-align:center;font-size:14px}
    body.dark-theme footer{background:#0a0a0a;color:var(--dt)}
    .social{display:flex;gap:15px;justify-content:center;margin-top:15px}
    .social a{display:flex;align-items:center;justify-content:center;width:40px;height:40px;background:rgba(255,255,255,.1);c
    .social a:hover{background:var(--la)}
    body.dark-theme .social a:hover{background:var(--da)}
    @media(max-width:768px){.nav{gap:15px;font-size:13px}h1{font-size:36px}h2{font-size:32px}.grid{grid-template-columns:1fr}.p
    @media(max-width:480px){.logo{font-size:20px}.nav{gap:10px}.hero{padding:40px 15px}h1{font-size:28px}h2{font-size:24px}.the
  </style>
ead>
dy class="light-theme">
  <header><div class="max"><div style="display:flex;justify-content:space-between;align-items:center;width:100%"><div class="logo
  <section class="hero" id="h"><div class="max"><h1>Your Files, Anywhere</h1><p>Secure cloud storage for all your devices</p><but
  <section class="features" id="f"><div class="max"><h2>Why CloudSync?</h2><div class="grid"><div class="card"><div class="icon">
  <section class="pricing"><div class="max"><h2>Pricing</h2><div class="grid"><div class="pcard"><h3>Basic</h3><p class="price">$
  <section class="contact" id="c"><div class="cform"><h2>Get In Touch</h2><form><input type="text" placeholder="Name" required><i
  <footer><div class="max"><p>&copy; 2024 CloudSync</p><div class="social"><a href="#" title="Facebook"><i class="fab fa-facebook
  <script>
    const b=document.getElementById('t'),d=document.body;
    function u(s){b.setAttribute('aria-pressed',s);b.querySelector('i').className=s?'fas fa-sun':'fas fa-moon';b.querySelector(
    function i(){const s=localStorage.getItem('theme')||(matchMedia('(prefers-color-scheme:dark)').matches?'dark':'light');d.cl
    b.onclick=()=>{const s=d.classList.contains('light-theme');d.className=(s?'dark':'light')+'-theme';u(s)};i();
  </script>
ody>
tml>
```

Output:-

In white Theme



In Black Theme



Justification:-

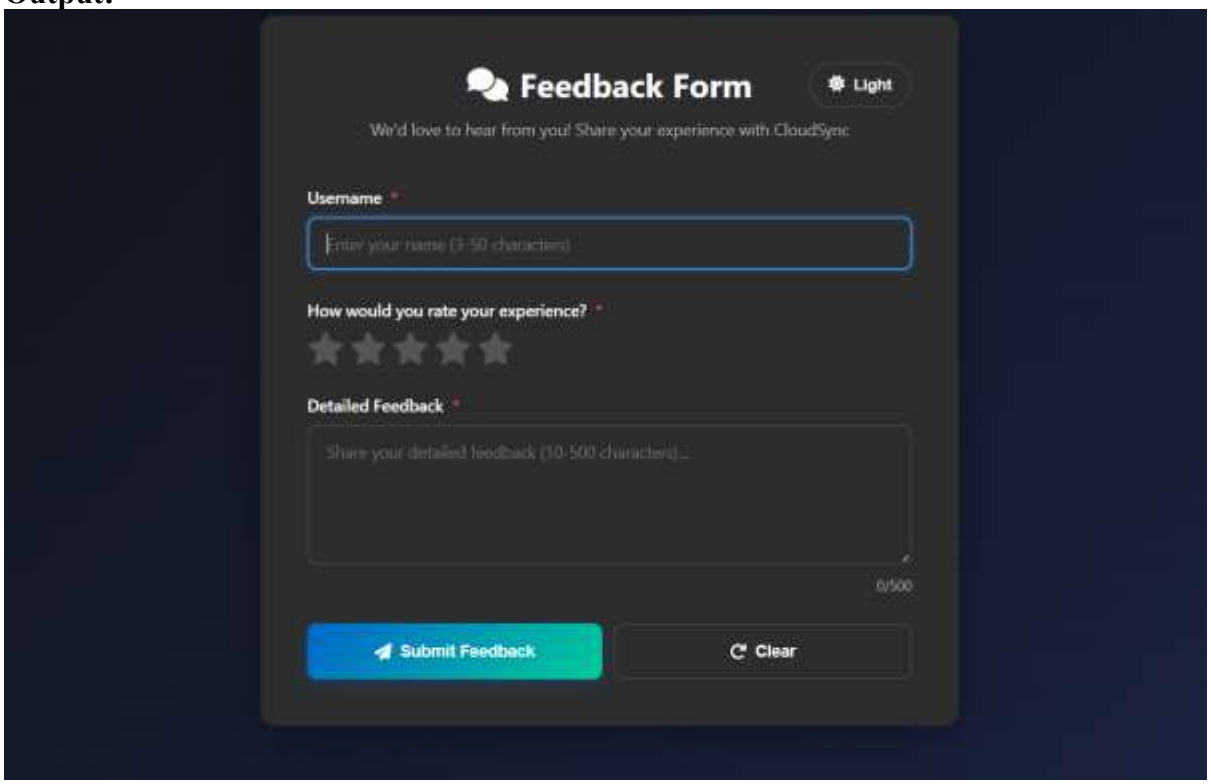
This prompt creates a theme toggle feature that improves user experience by allowing users to switch between light and dark modes. It uses JavaScript and CSS to update the theme instantly.

Task Description 3 – (User Feedback Form)
• Create a feedback form with fields: Username, Rating, Comments.
Requirements:
• Use JavaScript to validate required fields.
• Display success or error messages dynamically.
• Use AI to improve form usability and layout.

Prompt:-
Create a **feedback form** using HTML with fields for **Username, Rating, and Comments**. Use **JavaScript to validate required fields** and display a success or error message when the form is submitted.
Code:-

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>CloudSync - Feedback Form</title>
    <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0/css/all.min.css">
    <style>
        *{margin:0;padding:0;box-sizing:border-box}
        :root{--lb:#fff;--ls:#f5f5f5;--lt:#1a1a1a;--lm:#666;--la:#06c;--la2:#0c9;--ds:#2d2d2d;--dt:#fff;--dm:#b0b0b0;--d
        body{background:linear-gradient(135deg,#f5f7fa 0%,#c3cfe2 100%);font-family:Segoe UI,system-ui;min-height:100vh;padding:20px;tr
        body.dark-theme{background:linear-gradient(135deg,#0f1419 0%,#1a2342 100%)}
        .container{max-width:600px;margin:0 auto;background:var(--lb);padding:40px;border-radius:12px;box-shadow:0 10px 40px rgba(0,0,0,.
        body.dark-theme .container{background:var(--ds);box-shadow:0 10px 40px rgba(0,0,0,.3)}
        .header{text-align:center;margin-bottom:40px;position:relative}
        .header h1{color:var(--lt);font-size:28px;margin-bottom:10px;display:flex;gap:10px;align-items:center;justify-content:center}
        .header p{color:var(--lm);font-size:14px}
        body.dark-theme h1{color:var(--dt)}
        body.dark-theme .header p{color:var(--dm)}
        .theme-toggle{position:absolute;top:0;right:0;display:flex;gap:8px;padding:10px 16px;border:2px solid #e0e0e0;border-radius:25px;
        body.dark-theme .theme-toggle{background:var(--ds);color:var(--dt);border-color:#404040}
        .theme-toggle:hover{transform:translateY(-2px);border-color:var(--la);box-shadow:0 4px 16px rgba(0,102,204,.2)}
        body.dark-theme .theme-toggle:hover{border-color:var(--da);box-shadow:0 4px 16px rgba(51,153,255,.2)}
        .form-group{margin-bottom:25px}
        label{display:block;margin-bottom:8px;color:var(--lt);font-weight:600;font-size:14px}
        body.dark-theme label{color:var(--dt)}
        label .required{color:#e74c3c;margin-left:4px}
        input,textarea,select{width:100%;padding:12px 15px;border:2px solid #e0e0e0;border-radius:8px;font-family:inherit;font-size:14px;
        body.dark-theme input,body.dark-theme textarea,body.dark-theme select{background:var(--ds);color:var(--dt);border-color:#404040}
        input:focus,textarea:focus,select:focus{outline:0;border-color:var(--la);box-shadow:0 0 0 3px rgba(0,102,204,.1)}
        body.dark-theme input:focus,body.dark-theme textarea:focus,body.dark-theme select:focus{border-color:var(--da);box-shadow:0 0 0 3px
        input:valid{border-color:#27ae60}
        textarea{resize:vertical;min-height:120px}
        .star-rating{display:flex;gap:5px;margin-top:10px;line-height:1.6}
        .star{font-size:28px;color:#ddd;cursor:pointer;transition:.2s}
        body.dark-theme .star{color:#555}
        .star:hover,.star.active{color:#ffc107}
        .btn{flex:1;padding:14px 20px;border:none;border-radius:8px;font-weight:600;cursor:pointer;transition:all 0.3s;font-size:14px;displ
        .btn-submit{background:linear-gradient(135deg,var(--la),var(--la2));color:#fff;box-shadow:0 4px 15px rgba(0,102,204,.3)}
        .btn-submit:hover{transform:translateY(-2px);box-shadow:0 6px 20px rgba(0,102,204,.4)}
        body.dark-theme .btn-submit:hover{box-shadow:0 6px 20px rgba(51,153,255,.4)}
        .btn-reset{background:var(--ls);color:var(--lt);border:2px solid #e0e0e0}
        body.dark-theme .btn-reset{background:var(--ds);color:var(--dt);border-color:#404040}
        .btn-reset:hover{background:#e8e8e8}
        body.dark-theme .btn-reset:hover{background:#3a3a3a}
        .alert{padding:15px 20px;border-radius:8px;margin-bottom:25px;display:none;gap:12px;animation:slideDown 0.3s forwards;align-items:c
        .alert.show{display:flex}
        .alert i{font-size:20px}
        .alert-success{background:rgba(39,174,96,.1);color:#27ae60;border-left:4px solid #27ae60}
        body.dark-theme .alert-success{background:rgba(39,174,96,.15)}
        .alert-error{background:rgba(231,76,60,.1);color:#e74c3c;border-left:4px solid #e74c3c}
        body.dark-theme .alert-error{background:rgba(231,76,60,.15)}
        .field-error input,.field-error textarea,.field-error select{border-color:#e74c3c;background:rgba(231,76,60,.05)}
        body.dark-theme .field-error input,body.dark-theme .field-error textarea{background:rgba(231,76,60,.1)}
        .error-text{color:#e74c3c;font-size:12px;margin-top:6px;display:none;gap:4px;align-items:center}
```

Output:-



Justifications:-
This prompt generates a feedback form that collects user input and ensures required fields are filled before submission. It improves usability by guiding users and preventing incomplete form submissions.

Task Description - 4 (Image Gallery with JavaScript)
• Task: Display a small image gallery on a web page.
Requirements:
• Use JavaScript to dynamically load images.
• Add next/previous navigation buttons.
• Use AI to optimize layout responsiveness.

Prompt:-
Create a **simple image gallery** using **HTML, CSS, and JavaScript**. The gallery should load images dynamically and include **Next and Previous buttons** to navigate through the images.
code:

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```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Image Gallery</title>
    <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0/css/all.min.css">
    <style>
        *{margin:0;padding:0;box-sizing:border-box}
        :root{--lb:#fff;--lt:#1a1a1a;--lm:#666;--la:#06c;--db:#1a1a1a;--ds:#2d2d2d;--dt:#fff;--dm:#b0b0b0;--da:#39f}
        body{background:linear-gradient(135deg,#f5f7fa 0%,#c3cfe2 100%);font-family:Segoe UI,system-ui,sans-serif;min-height:100vh;padding:20px;transition:a
        body.dark-theme{background:linear-gradient(135deg,#0f141b 0%,#1a2342 100%)}
        .container{max-width:800px;width:100%;background:var(--lb);padding:30px;border-radius:12px;box-shadow:0 10px 40px rgba(0,0,0,.1);transition:all 0.3s}
        body.dark-theme .container{background:var(--ds);box-shadow:0 10px 40px rgba(0,0,0,.3)}
        .header{display:flex;justify-content:space-between;align-items:center;margin-bottom:30px;flex-wrap:wrap;gap:15px}
        .header h1{color:var(--lt);font-size:28px;transition:.3s}
        body.dark-theme .header h1{color:var(--dt)}
        .theme-toggle{display:flex;align-items:center;gap:8px;padding:10px 16px;border:2px solid #e0e0e0;border-radius:25px;background:var(--ls);color:var(--lt
        body.dark-theme .theme-toggle{background:var(--ds);color:var(--dt);border-color:#404040}
        .theme-toggle:hover{transform:translateY(-2px);box-shadow:0 4px 16px rgba(0,102,204,.25)}
        .gallery-container{position:relative;background:var(--lb);border-radius:12px;overflow:hidden;box-shadow:0 4px 15px rgba(0,0,0,.1);margin-bottom:20px;tr
        body.dark-theme .gallery-container{background:var(--ds);box-shadow:0 4px 15px rgba(0,0,0,.2)}
        .gallery-wrapper{position:relative;width:100%;padding-bottom:75%;overflow:hidden}
        .gallery-image{position:absolute;top:0;left:0;width:100%;height:100%;object-fit:cover;opacity:0;transition:opacity 0.5s ease-in-out}
        .gallery-image.active{opacity:1}
        .nav-button{position:absolute;top:50%;transform:translateY(-50%);background:rgba(0,102,204,.8);color:#fff;border:none;width:50px;height:50px;border-
        .nav-button:hover{background:rgba(0,102,204,1);transform:translateY(-50%) scale(1.1)}
        .nav-button.next{right:15px}
        .nav-button.prev{left:15px}
        .gallery-controls{display:flex;justify-content:space-between;align-items:center;padding:20px;background:var(--ls);border-radius:0 0 12px 12px;transition:
        body.dark-theme .gallery-controls{background:#1a1a1a}
        .image-counter{color:var(--lm);font-weight:600;text-align:center;flex:1;transition:.3s}
        body.dark-theme .image-counter{color:var(--dm)}
        .thumbnails{display:flex;gap:10px;justify-content:center;flex-wrap:wrap;margin-bottom:20px}
        .thumbnail{width:70px;height:70px;border-radius:8px;cursor:pointer;border:3px solid transparent;overflow:hidden;transition:all 0.3s;opacity:0.6}
        .thumbnail:hover,.thumbnail.active{opacity:1;border-color:var(--la);transform:scale(1.05)}
        body.dark-theme .thumbnail.active{border-color:var(--da)}
        .thumbnail img{width:100%;height:100%;object-fit:cover}
        .gallery-info{color:var(--lm);font-size:14px;text-align:center;margin-top:15px;transition:.3s}
        body.dark-theme .gallery-info{color:var(--dm)}
        @media(max-width:600px){.container{padding:20px}.header{flex-direction:column;gap:10px}.header h1{font-size:24px}.nav-button{width:40px;height:40px;font-
    </style>
</head>
<body class="light-theme">
    <div class="container">
        <div class="header">
            <h1><i class="fas fa-images"></i> Photo Gallery</h1>
            <button id="themeToggle" class="theme-toggle" aria-pressed="false">
                <i class="fas fa-moon"></i><span>Dark</span>
            </button>
        </div>

        <div class="gallery-container">
            <div class="gallery-wrapper" id="galleryWrapper">
```

```
    <div class="gallery-info">
        <i class="fas fa-arrow-left"></i> Use arrow buttons or click thumbnails to navigate <i class="fas fa-arrow-right"></i>
    </div>
</div>

<script>
    const images=[
        {src:'https://images.unsplash.com/photo-1506905925346-21bda4d32df4?w=800&h=600&fit=crop',title:'Nature Valley'},
        {src:'https://images.unsplash.com/photo-1506905925346-21bda4d32df4?w=800&h=600&fit=crop&dpr=2&auto=format,https://images.unsplash.com/photo-1519904981063-b0ff2531823f',title:'Mountain View'},
        {src:'https://images.unsplash.com/photo-1441716844725-09cedc13a4bf?w=800&h=600&fit=crop',title:'Sunset Beach'},
        {src:'https://images.unsplash.com/photo-1447933601403-0c6688de566e?w=800&h=600&fit=crop',title:'Forest Path'},
        {src:'https://images.unsplash.com/photo-1506905925346-21bda4d32df4?w=800&h=600&fit=crop',title:'Blue Sky'},
        {src:'https://images.unsplash.com/photo-1519904981063-b0ff2531823f?w=800&h=600&fit=crop',title:'Snow Peak'}
    ];

    let current=0;
    const body=document.body;
    const gallery=document.getElementById('gallery-grid');
    const counter=document.querySelector('.image-counter');
    const prevBtn=document.getElementById('prevBtn');
    const nextBtn=document.getElementById('nextBtn');
    const themeToggle=document.getElementById('themeToggle');
    const thumbnailsContainer=document.getElementById('thumbnails');

    // Load images dynamically
    function loadGallery(){
        gallery.innerHTML=images.map((img,i)=>`<img class="gallery-image ${i===0?'active':''}" src="${img.src}" alt="${img.title}">`).join('');
        thumbnailsContainer.innerHTML=images.map((img,i)=>`<div class="thumbnail ${i===0?'active':''}" onclick="goToImage(${i})" title="${img.title}"><img src="${img.src}" alt="${img.title}"></div>`).join('');
        updateCounter();
    }

    function updateCounter(){counter.innerHTML= <i class="fas fa-photo-video"></i> Image ${current+1} of ${images.length} }

    function showImage(index){
        document.querySelectorAll('.gallery-image').forEach(img=>img.classList.remove('active'));
        document.querySelectorAll('.thumbnail').forEach(thumb=>thumb.classList.remove('active'));
        document.querySelectorAll('.gallery-image')[index].classList.add('active');
        document.querySelectorAll('.thumbnail')[index].classList.add('active');
        updateCounter();
    }

    function nextImage(){current=(current+1)%images.length;showImage(current)}
    function prevImage(){current=(current-1+images.length)%images.length;showImage(current)}
    function goToImage(index){current=index;showImage(current)}

    // Theme toggle
    function initTheme(){const t=localStorage.getItem('theme')||(matchMedia('(prefers-color-scheme:dark)').matches?'dark':'light');body.classList.toggle('light-theme',t==='light');updateThemeUI(t==='dark')}
    function updateThemeUI(isDark){themeToggle.setAttribute('aria-pressed',isDark);themeToggle.querySelector('i').classList.toggle(isDark?'fas fa-sun':'fas fa-moon');themeToggle.querySelector('span').textContent=isDark?
    themeToggle.onclick=()=>{const isDarkNow=!body.classList.contains('light-theme');body.classList.toggle('light-theme',isDarkNow);localStorage.setItem('theme',isDarkNow?'dark':'light');updateThemeUI(isDarkNow)};

    // Keyboard navigation
    document.addEventListener('keydown',e=>{if(e.key==='ArrowRight')nextImage();if(e.key==='ArrowLeft')prevImage()});

    // Event listeners
    nextBtn.onclick=nextImage;
    prevBtn.onclick=prevImage;

    // Initialize
    loadGallery();
    initTheme();
</script>
</body>
```

Output:-

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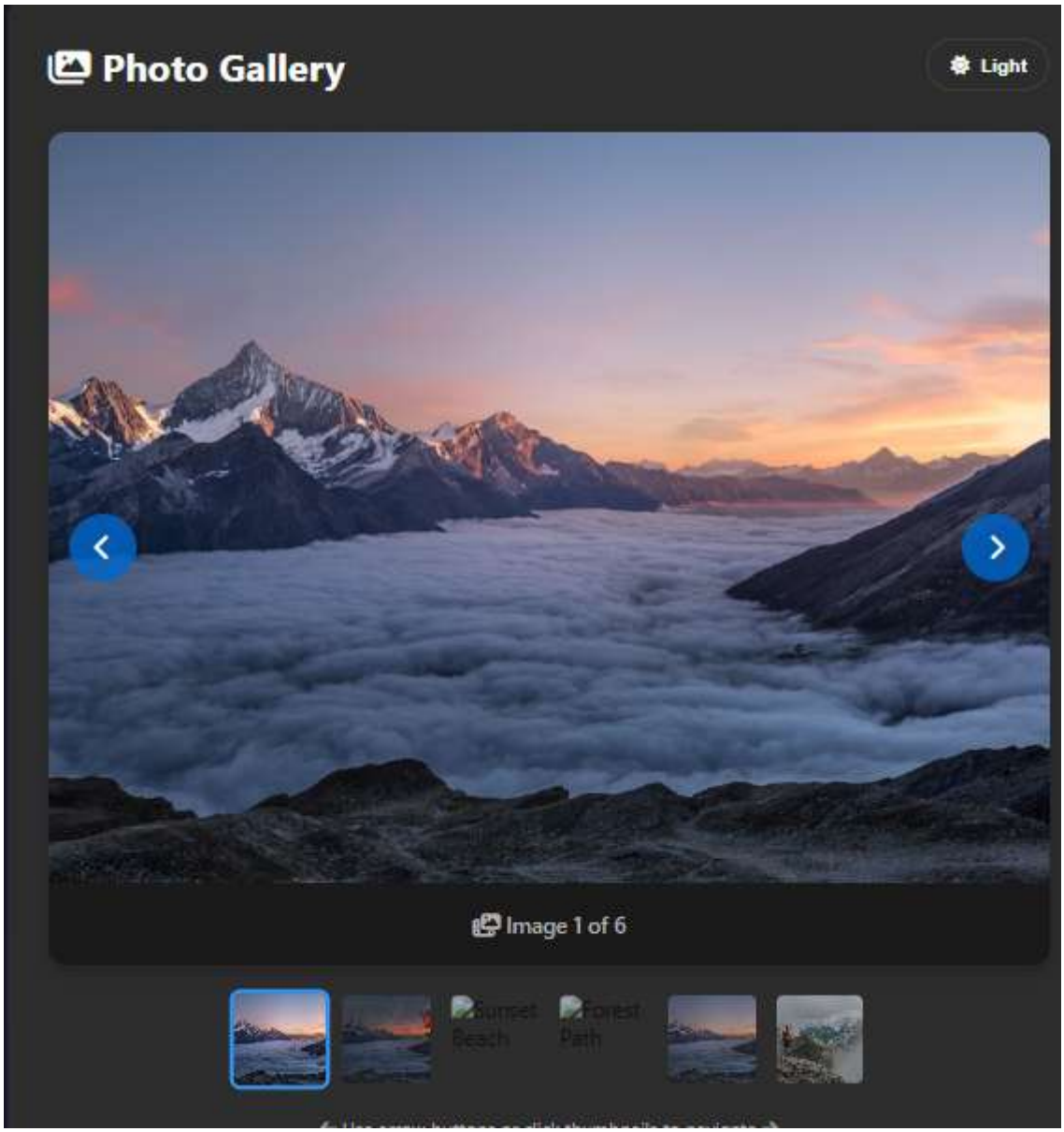

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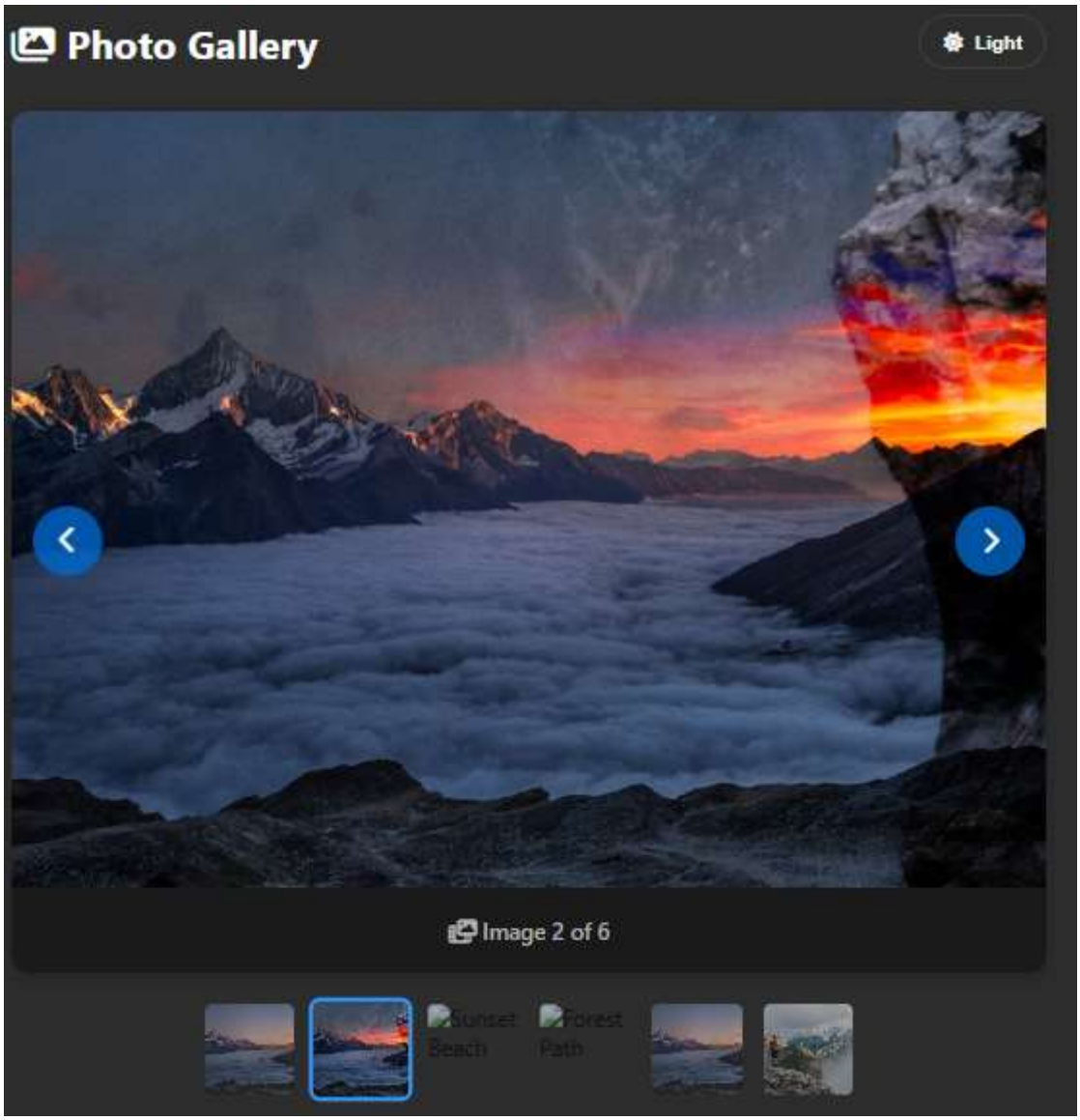

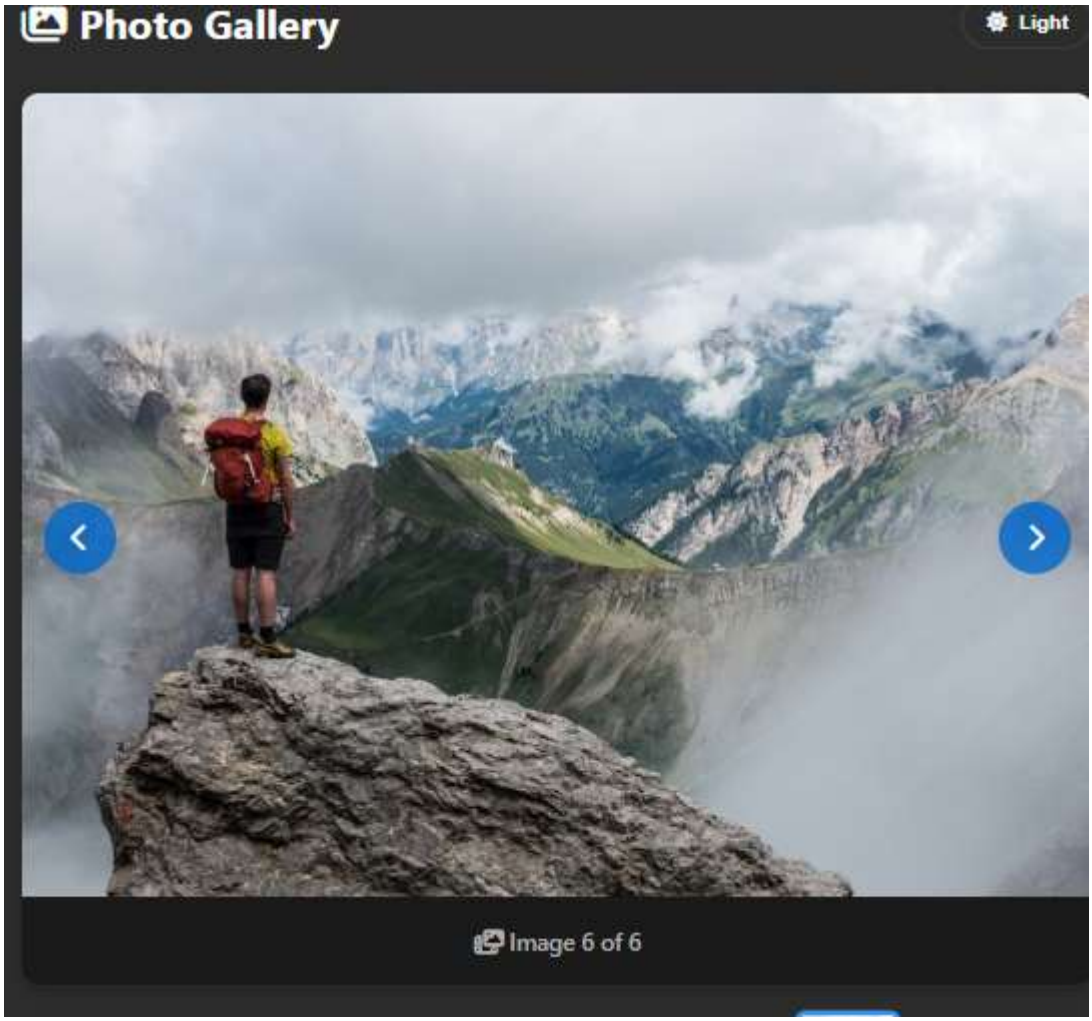

Justification:-
This prompt builds an interactive image gallery where users can easily browse images using navigation buttons. It improves the visual experience and keeps the layout responsive.

-
Task Description - 5: (Notification Banner)
• Display a notification banner at the top of the page.
Requirements:
• Use JavaScript to show and hide the banner.
• Style the banner using CSS.
• Use AI to ensure smooth transitions and clean UI behavior.

Prompt:-
Create a **notification banner at the top of a webpage** using HTML and CSS. Use **JavaScript to show and hide the banner** when the user clicks a button.

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Code:

-

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Notification Banner</title>
    <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.4.0/css/all.min.css">
    <style>
        *{margin:0;padding:0;box-sizing:border-box}
        :root{--bg:#fff;--text:#333;--bg-dark:#1a1a1a;--text-dark:#e0e0e0;--primary:#007bff;--success:#28a745;--warning:#ffc107;--danger:#dc3545;--info:#17a2b8}
        body{background:linear-gradient(135deg,#667eea 0%,#764ba2 100%);font-family:Segoe UI,system-ui,sans-serif;min-height:100vh;padding:20px;transition:all 0.3s}
        body.dark-theme{background:linear-gradient(135deg,#0f2027 0%,#203a43 50%,#2c5364 100%)}
        .notifications-container{position:fixed;top:20px;right:20px;z-index:9999;max-width:400px}
        .notification{background:var(--bg);padding:15px 20px;border-radius:8px;margin-bottom:12px;display:flex;align-items:center;gap:12px;box-shadow:0 4px 20px rgba(0,0,0,.15);animation:slideIn 0.3s ease-out,sli
        body.dark-theme .notification{background:var(--bg-dark);box-shadow:0 4px 20px rgba(0,0,0,.3)}
        .notification.hide{animation:slideOut 0.3s ease-out forwards}
        @keyframes slideIn{from{transform:translateX(400px);opacity:0}to{transform:translateX(0);opacity:1}}
        @keyframes slideOut{to{transform:translateX(400px);opacity:0}}
        .notification.success{border-left-color:#27ae60;background:rgba(39,174,96,.05)}
        body.dark-theme .notification.success{background:rgba(39,174,96,.1)}
        .notification.error{border-left-color:#e74c3c;background:rgba(231,76,60,.05)}
        body.dark-theme .notification.error{background:rgba(231,76,60,.1)}
        .notification.warning{border-left-color:#f39c12;background:rgba(243,156,18,.05)}
        body.dark-theme .notification.warning{background:rgba(243,156,18,.1)}
        .notification.info{border-left-color:#3498db;background:rgba(52,152,219,.05)}
        body.dark-theme .notification.info{background:rgba(52,152,219,.1)}
        .notification.success .icon{color:#27ae60}
        .notification.error .icon{color:#e74c3c}
        .notification.warning .icon{color:#f39c12}
        .notification.info .icon{color:#3498db}
        .icon{font-size:24px;flex-shrink:0}
        .notification-content{flex:1;color:var(--text)}
        body.dark-theme .notification-content{color:var(--text)}
        .notification-title{font-weight:600;font-size:15px}
        .notification-message{font-size:13px;color:var(--text);margin-top:5px}
        body.dark-theme .notification-message{color:var(--text)}
        .notification-close{background:none;border:none;color:var(--text);cursor:pointer;font-size:18px;padding:0;transition:.2s}
        body.dark-theme .notification-close{color:var(--text)}
        .notification-close:hover{transform:scale(1.2);color:var(--text)}
        body.dark-theme .notification-close:hover{color:var(--text)}
        .progress-bar{position:absolute;bottom:0;left:0;height:3px;background:inherit;animation:progress 8s linear forwards}
        .notification.success .progress-bar{background:#27ae60}
        .notification.error .progress-bar{background:#e74c3c}
        .notification.warning .progress-bar{background:#f39c12}
        .notification.info .progress-bar{background:#3498db}
        @keyframes progress{to{width:0}}
        .container{max-width:900px;margin:0 auto;background:var(--bg);padding:40px;border-radius:12px;box-shadow:0 10px 40px rgba(0,0,0,.1);transition:all 0.3s}
        body.dark-theme .container{background:var(--text);box-shadow:0 10px 40px rgba(0,0,0,.3)}
        .header{display:flex;justify-content:space-between;align-items:center;margin-bottom:40px;flex-wrap:wrap;gap:20px}
        .header h1{color:var(--text);font-size:32px;transition:.3s}
        body.dark-theme .header h1{color:var(--text)}
        .theme-toggle{display:flex;align-items:center;gap:8px;padding:10px 15px;border:2px solid #e0e0e0;border-radius:25px;background:#f5f5f5;color:var(--bg);font-weight:600;cursor:pointer;transition:.3s;font-
        body.dark-theme .theme-toggle{background:var(--text);color:var(--text);border-color:#444444}
        .theme-toggle:hover{transform:translateY(-2px);box-shadow:0 4px 15px rgba(0,102,204,.25)}
        .demo-section{margin-bottom:40px}
        .demo-section h2{color:var(--text);font-size:20px;margin-bottom:15px;transition:.3s}
        body.dark-theme .demo-section h2{color:var(--text)}
        .btn{padding:12px 20px;border:none;border-radius:8px;font-weight:500;cursor:pointer;transition:all 0.3s;font-size:14px;display:inline-flex;align-items:center;gap:8px;margin-right:10px;margin-bottom:10px}
        .btn.success{background:#27ae60;color:#fff}
        .btn.success:hover{transform:translateY(-2px);box-shadow:0 4px 15px rgba(39,174,96,.3)}
        .btn.error{background:#e74c3c;color:#fff}
        .btn.error:hover{transform:translateY(-2px);box-shadow:0 4px 15px rgba(231,76,60,.3)}
        .btn.warning{background:#f39c12;color:#fff}
        .btn.warning:hover{transform:translateY(-2px);box-shadow:0 4px 15px rgba(243,156,18,.3)}
        .btn.info{background:#3498db;color:#fff}
        .btn.info:hover{transform:translateY(-2px);box-shadow:0 4px 15px rgba(52,152,219,.3)}
        .code-block{background:var(--bg);color:#fff;padding:15px;border-radius:8px;font-family:monospace;font-size:12px;overflow-x:auto;margin-top:15px;transition:.3s}
```

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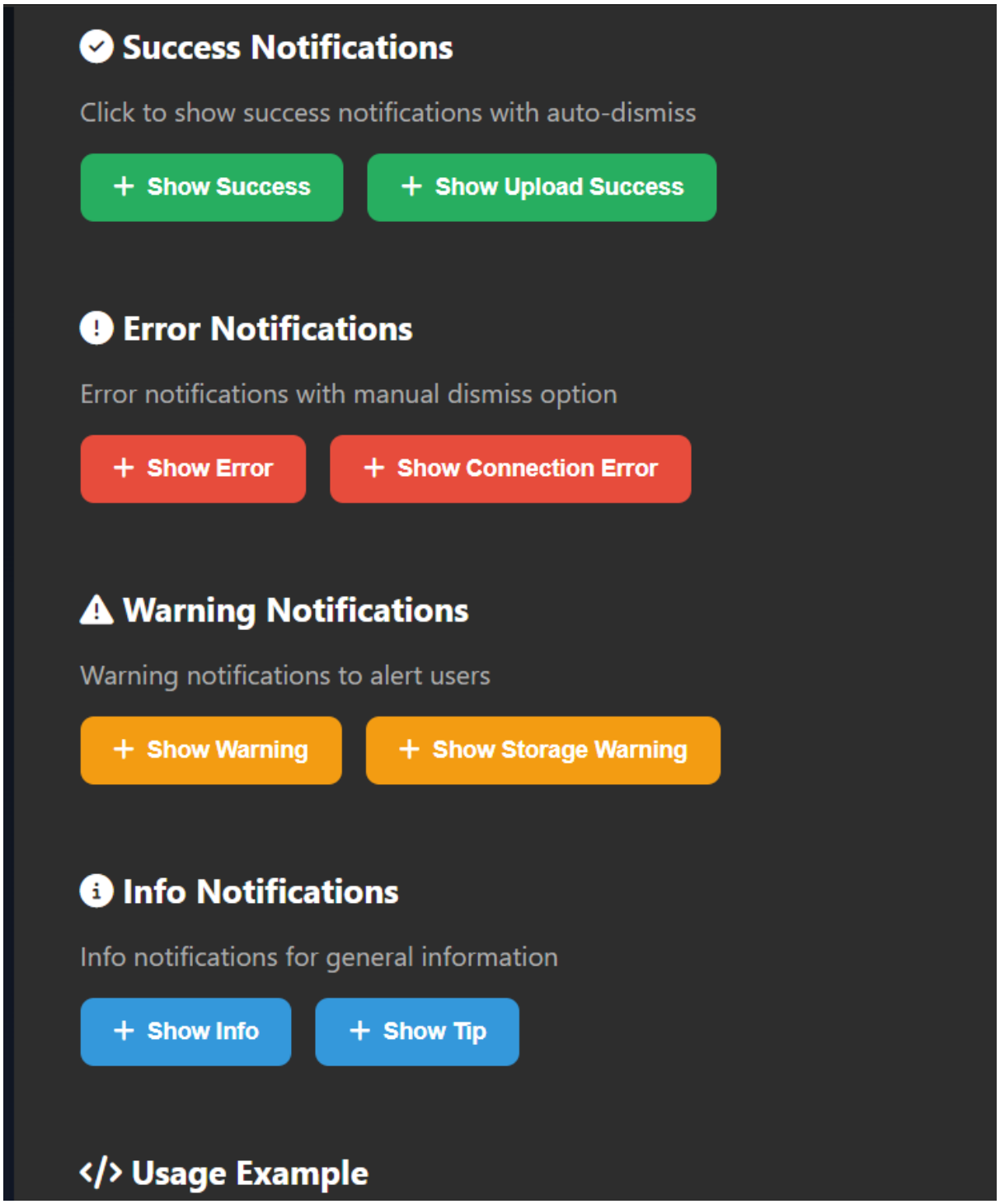
```
        .btn-info:hover{transform:translateY(-2px);box-shadow:0 4px 15px rgba(52,152,219,.3)}
        .code-block{background:var(--lm);color:#fff;padding:15px;border-radius:8px;font-family:monospace;font-size:12px;overflow-x:auto;margin-top:15px;tran
        body.dark-theme .code-block{background:#111}
        .section-description{color:var(--lm);margin-bottom:15px;transition:.3s}
        body.dark-theme .section-description{color:var(--dm)}
        @media(max-width:600px){.notifications-container{max-width:100%;right:10px;left:10px}.container{padding:20px}.header{flex-direction:column;gap:15px}.he
    </style>
</head>
<body class="light-theme">
    <div class="notifications-container" id="notificationsContainer"></div>

    <div class="container">
        <div class="header">
            <h1><i class="fas fa-bell"></i> Notification Demo</h1>
            <button id="themeToggle" class="theme-toggle" aria-pressed="false">
                <i class="fas fa-moon"></i><span>Dark</span>
            </button>
        </div>

        <div class="demo-section">
            <h2><i class="fas fa-check-circle"></i> Success Notifications</h2>
            <p class="section-description">Click to show success notifications with auto-dismiss</p>
            <button class="btn btn-success" onclick="notify('success','Operation Successful!','Your changes have been saved.','fa-check-circle')">
                <i class="fas fa-plus"></i> Show Success
            </button>
            <button class="btn btn-success" onclick="notify('success','File Uploaded','image.jpg (2.5 MB) uploaded successfully.','fa-cloud-upload-alt')">
                <i class="fas fa-plus"></i> Show Upload Success
            </button>
        </div>

        <div class="demo-section">
            <h2><i class="fas fa-exclamation-circle"></i> Error Notifications</h2>
            <p class="section-description">Error notifications with manual dismiss option</p>
            <button class="btn btn-error" onclick="notify('error','Error Occurred!','Something went wrong. Please try again.','fa-exclamation-circle')">
                <i class="fas fa-plus"></i> Show Error
            </button>
            <button class="btn btn-error" onclick="notify('error','Connection Failed','Unable to reach the server.','fa-wifi')">
                <i class="fas fa-plus"></i> Show Connection Error
            </button>
        </div>

        <div class="demo-section">
            <h2><i class="fas fa-exclamation-triangle"></i> Warning Notifications</h2>
            <p class="section-description">Warning notifications to alert users</p>
            <button class="btn btn-warning" onclick="notify('warning','Warning!','This action cannot be undone.','fa-exclamation-triangle')">
                <i class="fas fa-plus"></i> Show Warning
            </button>
            <button class="btn btn-warning" onclick="notify('warning','Low Storage','You have less than 1GB available.','fa-hard-drive')">
                <i class="fas fa-plus"></i> Show Storage Warning
            </button>
        </div>

        <div class="demo-section">
            <h2><i class="fas fa-info-circle"></i> Info Notifications</h2>
            <p class="section-description">Info notifications for general information</p>
            <button class="btn btn-info" onclick="notify('info','New Update','Version 2.0 is now available.','fa-info-circle')">
                <i class="fas fa-plus"></i> Show Info
            </button>
            <button class="btn btn-info" onclick="notify('info','Tip','Use keyboard shortcuts for faster navigation.','fa-lightbulb')">
                <i class="fas fa-plus"></i> Show Tip
            </button>
        </div>
```

```
        </div>

        <div class="demo-section">
            <h2><i class="fas fa-code"></i> Usage Example</h2>
            <p class="section-description">Simple JavaScript function to create notifications</p>
            <div class="code-block">
notify('success', 'Title', 'Message', 'icon-name', 3000);
// Types: 'success', 'error', 'warning', 'info'
// Duration: auto-dismiss time in milliseconds (0 = manual close only)
            </div>
        </div>
    </div>

    <script>
        const body=document.body;
        const themeToggle=document.getElementById('themeToggle');
        const container=document.getElementById('notificationsContainer');

        // Theme toggle
        function initTheme(){const t=localStorage.getItem('theme')||(matchMedia('(prefers-color-scheme:dark)').matches?'dark':'light
        function updateThemeUI(isDark){themeToggle.setAttribute('aria-pressed',isDark);themeToggle.querySelector('i').className=isDa
        themeToggle.onclick=()=>{const isDark=body.classList.contains('light-theme');body.className=(isDark?'dark':'light')+'-theme'

        // Notification system
        function notify(type,title,message,icon='fa-bell',duration=3000){
            const notif=document.createElement('div');
            const id='notif-'+Date.now();
            notif.id=id;
            notif.className=`notification ${type}`;
            notif.innerHTML=`
                <i class="fas ${icon} icon"></i>
                <div class="notification-content">
                    <div class="notification-title">${title}</div>
                    <div class="notification-message">${message}</div>
                </div>
                <button class="notification-close" onclick="removeNotif('${id}')" aria-label="Close notification">
                    <i class="fas fa-times"></i>
                </button>
                ${duration>0?`<div class="progress-bar" style="animation-duration:${duration}ms"></div>`:''}
            `;
            container.appendChild(notif);
            if(duration>0)setTimeout(()=>removeNotif(id),duration);
        }

        function removeNotif(id){
            const notif=document.getElementById(id);
            if(notif){
                notif.classList.add('hide');
                setTimeout(()=>notif.remove(),300);
            }
        }

        // Show welcome notification
        window.addEventListener('load',()=>{
            notify('info','Welcome!','Click buttons below to see different notification types.','fa-hand-sparkles',4000);
        });

        initTheme();
    </script>
</body>
</html>
```

Output:-



Justification:-
Displays a notification banner at the top of the page that can be shown or hidden using JavaScript. Uses smooth transitions and styled layout to deliver important messages without disrupting the overall UI.